

Therapeutic Vaccines

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Vaccines and Unmet Medical Needs

- Infectious diseases were historically a major cause of death.
- Vaccination has greatly reduced morbidity and mortality from diseases such as:
 - smallpox
 - poliomyelitis
 - diphtheria
 - tetanus
 - Measles
- COVID-19 vaccines showed high efficacy in preventing severe disease
- However, infectious diseases still remain a major global health burden.

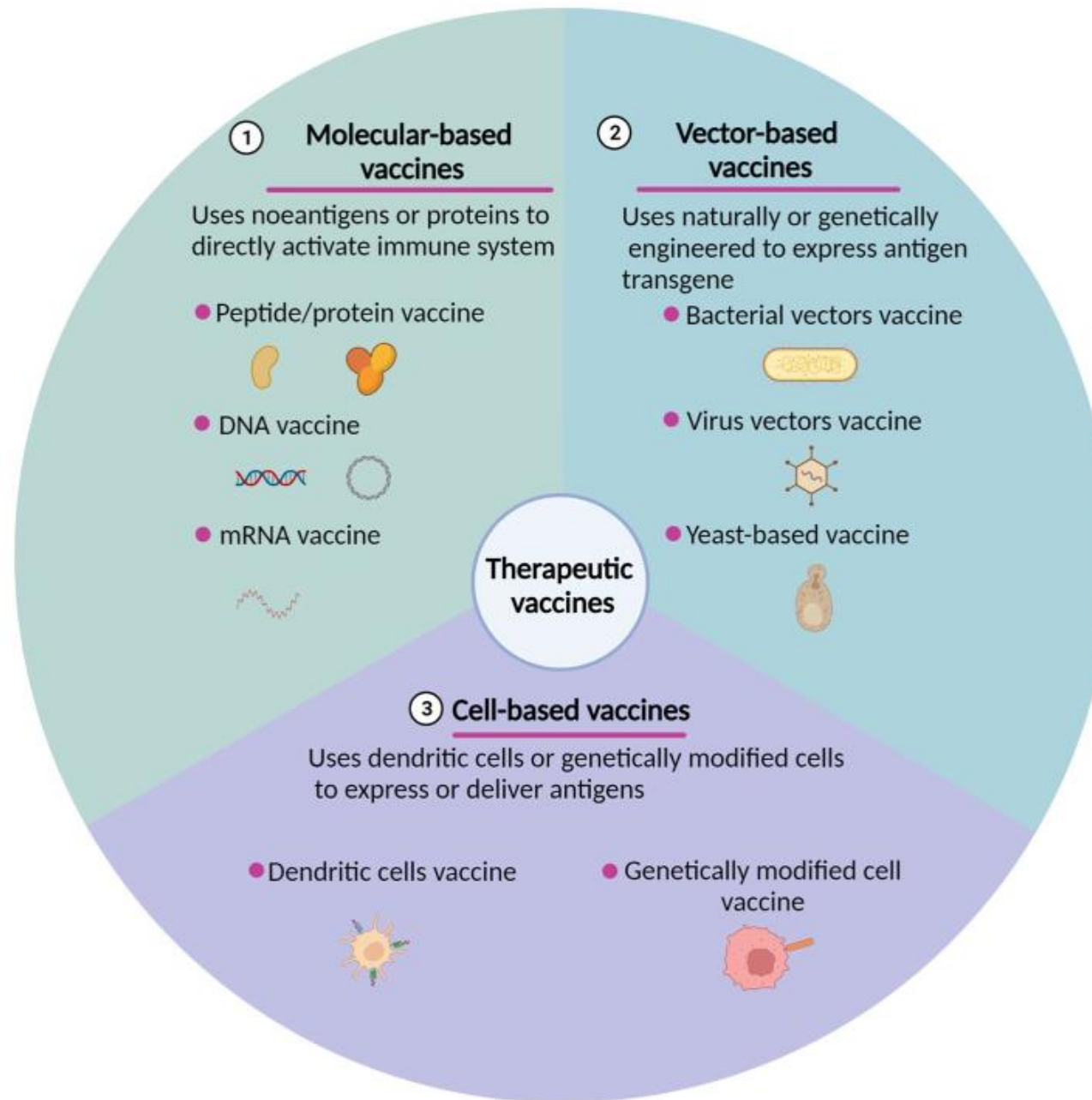
Why Preventive Vaccines Are Not Enough

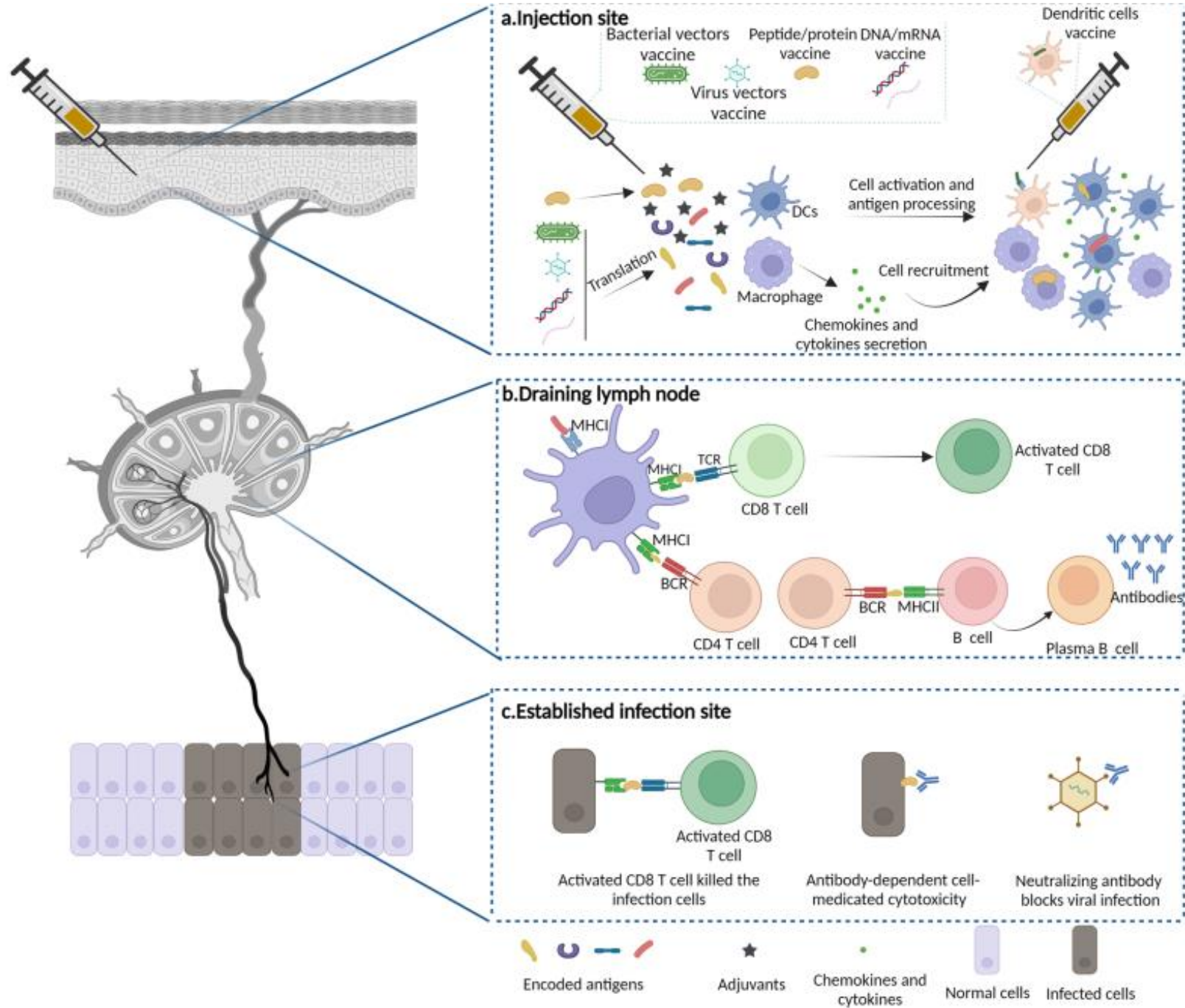
- Despite highly effective COVID-19 vaccines, many individuals still developed SARS-CoV-2 infection.
- Key limitations include:
 - limited vaccine acceptance
 - inequitable global distribution
 - short duration of protection
 - limited cross-protection
 - viral variants
 - immunosuppression in some individuals
- Other infectious diseases such as HPV, HBV, HIV, HCV, and influenza continue to cause mortality.

Therapeutic Vaccine Concept: Targeting Persistent/Unique Antigens

- Established diseases often show **constant or unique antigen expression**.
- Example targets:
 - **HPV E6/E7**
 - **tumor neoantigens**
 - **HBsAg (HBV)**
- Goal: induce **epitope-specific effector T cells** and/or **neutralizing antibodies**.
- Preclinical proof-of-concept:
 - **E7 protein vaccine** induced strong cellular immunity → **rejection of established tumors**

An overview of therapeutic vaccine types in preclinical and clinical trials.





Immunologic Basis of Therapeutic HPV Vaccination

Therapeutic HPV vaccines aim to induce:

- **strong Th1-type responses**
- **cytotoxic CD8+ T-cell responses**
- Goal: eliminate **HPV-infected** and **transformed cells**
- An ideal target antigen should be:
 - essential for malignant transformation and maintenance
 - constitutively expressed
 - highly expressed
 - minimally mutable
- **HPV E6 and E7 oncoproteins** are near-ideal targets
- Therefore, most HPV therapeutic vaccines are designed around **E6/E7**

Vector-Based HPV Therapeutic Vaccines

Bacterial vectors

- Examples: **Listeria monocytogenes**, **Lactobacillus casei**, **Lactococcus/Lactobacillus**, **Salmonella**
- Advantages:
 - intrinsic adjuvant activity
 - MHC I / MHC II antigen presentation
 - activation of **CD8+** and **CD4+** T cells

Key examples

- **ADX11-001**: attenuated *Listeria* expressing **HPV16 E7-LLO**
 - acceptable safety in phase I
 - induced HPV16 E7-specific T-cell responses
 - phase II showed improved 12-month survival in advanced cervical cancer
 - phase III ongoing
- **GLBL101c / NZ8123-HPV16-optiE6**
 - induced mucosal or E6-specific immune responses
 - showed encouraging early clinical activity

Viral vectors

- Platforms: **MVA**, **adenovirus**, **chimpanzee adenovirus**, **Semliki Forest virus**, **arenavirus**
- **TA-HPV** and **Tipapkinogen Sovacivec**
 - safe, immunogenic
 - induced HPV-specific CTLs

Peptide- and Protein-Based HPV Vaccines

Peptide vaccines

- Advantages:

- easy to manufacture
- epitope-specific

- Limitation:

- low intrinsic immunogenicity
- usually require **adjuvants** (e.g. cytokines, TLR ligands, Montanide)

Key example: ISA101

- Synthetic long-peptide vaccine targeting **HPV16 E6/E7**
- Response correlated with stronger **CD4+ / CD8+ IFN- γ** responses
- Combination with **nivolumab** showed promising survival data in incurable HPV16 cancer

Protein vaccines

- Broader epitope content than peptides
- Often more humoral-biased unless specifically engineered
- Examples:
 - **GTL001**
 - **SCN-00101**
 - **TVGV-1**
- Fusion strategies enhance APC targeting and immunogenicity

DNA Vaccines and Clinical Outlook in HPV

- **VGX-3100** (HPV16/18 E6/E7 DNA vaccine)

- phase I: strong **Th1-biased** responses
- phase II: higher histopathologic regression than placebo

- **GX-188E** fused with **Flt3L**

- to enhance antigen presentation
- induced strong polyfunctional **E6/E7-specific T-cell responses**
- associated with HPV clearance and CIN3 regression

Other candidates: **pNGVL4a-CRT-E7**, **ZYC101**

Viral Vector Platforms for HBV Immunotherapy

- **Adenoviral Vectors:** Strong inducer of innate signaling pathways; promotes robust humoral and cellular responses.

- **TG1050 Vaccine:**

- Non-replicative adenoviral vector encoding HBV core, polymerase, and envelope domains.
- Preclinical: Activated T-cell cytolytic activity and showed antiviral effects.
- Phase I Trial (NCT02428400): Well-tolerated, induced IFN- γ -producing T cells, and synergized with NUC therapy.

- **Heterologous Prime-Boost Strategies:**

- Chimpanzee Adenovirus (Prime) followed by **MVA** (Boost) encoding multiple HBV antigens.
- Co-ordinated enhancement of polyfunctional HBV-specific CD8⁺ and CD4⁺ T-cell responses in vivo.

- **Gene Editing Delivery:** Utilizing adenovirus vectors to deliver CRISPR/Cas9 systems targeting the HBV genome.

Protein- and Peptide-Based HBV Vaccines

- Theravax (DV-601):** Recombinant HBsAg and HBcAg with saponin-based ISCOMATRIX adjuvant; induced lymphoproliferative and T-cell responses.
- GS-4774:** Engineered yeast expressing an HBsAg-HBcAg-HBx fusion.
 - Yeast cell wall acts as a strong adjuvant for DC presentation.
 - Phase II trials showed GS-4774 was safe and boosted CD8+ T cells, but did not significantly reduce serum HBsAg alone.
- HeberNasvac (Approved Candidate):**
 - Contains both HBsAg and HBcAg (nasal/subcutaneous delivery).
 - Phase III trial vs. Peg-IFN: Induced superior reduction in HBV DNA load (undetectable levels).
 - Approved by the Cuban Regulatory Authority (2015); evaluated as adjunct to NUCs (NCT02249988).

HepTcell: 9 synthetic CD4+/CD8+ peptides derived from conserved HBV domains + IC31 adjuvant.

DNA-Based HBV Vaccines: Adjuvants & Synergy

- Mechanism:** Plasmids encoding HBV antigens directly transfect host cells to drive endogenously processed antigen expression.

- Key Candidates:**

- INO-1800:** Synthetic DNA vaccine encoding HBsAg and consensus HBcAg.

- INO-9112 (Co-delivery):** Plasmid expressing **IL-12** co-administered with INO-1800 to boost cytotoxic T lymphocyte (CTL) activation.

- HB-110:** Multivalent plasmid encoding HBs, PreS1, HBc, Hbpol, and IL-12; showed strong antibody and T-cell responses in preclinical models.

Outlook: DNA platforms benefit from manufacturing ease and the ability to easily co-deliver genetic adjuvants (like IL-12) to rescue exhausted T cells.

Peptide, Protein, & Yeast-Based Platforms for HCV

•Peptide/Protein Vaccines:

- Relatively easy to manufacture but often suffer from HLA-restriction and low immunogenicity.
- **IC41 Vaccine:** 5 synthetic peptides (Core, NS3, NS4) + poly-L-arginine adjuvant.
- *Phase I (HLA-A2+)*: Safe and immunogenic, but T-cell responses were too weak to clear viral RNA.
- *Intensified/Intradermal Dosing*: Showed improved, more robust immune responses.
 - **HCV Core (C35–44) + ISA51**: Safe and well-tolerated in Phase I.
 - **NCT00445419**: Virosome-based vaccine containing NS3 peptides (Phase I completed).

•Yeast Vector Vaccine:

- **GI-5005**: Yeast-vector expressing an **NS3-Core fusion protein**.
- *Preclinical*: Induced robust NS3- and core-specific cellular responses.
- *Phase I (NCT00124215)*: Safe, well-tolerated, and induced significant HCV-specific immunity.
- *Phase II (NCT00606086)*: Evaluated combined efficacy with standard-of-care

DNA-Based & Recombinant Viral Vector Vaccines for HCV

- **DNA Vaccines:** Designed to express structural/non-structural proteins (NS3, NS4, NS5, Core, E1/E2) to trigger both humoral and cellular immunity.
 - **CIGB-230:** First therapeutic DNA vaccine evaluated in clinical trials; contains a mixture of core/E1/E2-expressing plasmids.
 - **ChronVac-C (NCT01335711):** Phase II trial; encodes conserved NS3/NS4a regions. Initial data confirmed safety and immunogenicity.
 - **Modified NS3 DNA Vaccines:** Engineered mutations in the catalytic triad/helicase domains to prevent the vaccine antigen from interfering with normal host cell functions.
- **Viral Vector-Based Vaccines:**
 - **Ad6-based vaccine:** Encodes NS3, NS4A, NS4B, NS5A, and inactive NS5B.
 - *Trial (NCT01094873):* Safe and highly immunogenic in Phase I (previously validated in mice, macaques, and chimpanzees).
 - **TG4040 (MVA-based):** Expresses NS3/4/5B. Induces potent, long-lasting T-cell responses. Evaluated in Phase II (NCT01055821) in combination with PEG-IFN/Ribavirin.
 - **Dendritic Cell (DC) Vaccines for HCV:** Autologous DCs loaded and activated *ex vivo* with HCV HLA-A2 restricted T-cell epitopes.
 - **Clinical Trial Results:** Tested in chronic HCV patients; showed safety but only weak HCV-specific CD8+ T-cell responses.

Table 1 Selected clinical trials of therapeutic vaccines against infectious disease

Therapeutic HPV vaccines					
Vaccine type	Vaccine Formulation	Combination agents	Condition	Clinical trial identifier	Phase/status
Bacterial vector vaccine	ADXS11-001 (attenuated live <i>Listeria</i> Encoding HPV 16 E7 vector)	Placebo	Cervical carcinoma	NCT02164461	Phase I-II Completed
			Cervical carcinoma Head and Neck Cancer	NCT02291055	Phase I-II Active, not recruiting
			Cervical cancer	NCT01266460	Phase II Completed
			Cervical cancer	NCT02853604	Phase III Active, not recruiting
			Anal cancer Rectal cancer	NCT02399813	Phase II Completed
			Head and neck cancer, oropharyngeal squamous cell carcinoma	NCT02002182	Phase II Active, not recruiting
Viral vector vaccine	Vax001 (Semliki Forest virus expressing E6 and E7)	Nivolumab	CIN 2/3 Cervical cancer	NCT03141463	Phase I Completed
	HB-201 (lymphocytic choriomeningitis virus encoding HPV E6 and E7)		HPV-Related Squamous Cell Carcinoma	NCT04180215	Phase I/II Recruiting
Peptide/protein vaccine	ISA101(nine HPV-16 E6 and four HPV-16 E7 synthetic peptides with Montanide ISA51)	Nivolumab	Solid Tumors	NCT02426892	Phase II Completed
	TVGV-1 (HPV16 E7 fused protein +adjuvant GPI-0100)		High-grade squamous intraepithelial lesions	NCT02576561	Phase IIa Active, not recruiting
DNA vaccine	VGX-3100(plasmid encoding E6 and E7 of HPV16/18)	Placebo	Head and neck squamous cell cancer	NCT02163057	Phase I/IIa Completed
			Cervical cancer	NCT02172911	Phase I/IIa Completed
			High-grade squamous intraepithelial lesion of the cervix	NCT03185013	Phase III Completed
	GX188E (HPV E6/E7 fused to Flt3L)	Placebo	Cervical intraepithelial neoplasia	NCT02139267	Phase II Completed
			Cervical Intraepithelial Neoplasia	NCT02596243	Phase II Active, not recruiting
			Cervical Intraepithelial Neoplasia 3	NCT03206138	Phase Ib/II Recruiting
Therapeutic HBV vaccines					
Viral vector vaccines	TG1050(Adeno vector encoding core, polymerase, envelope fusion protein)	Placebo	Chronic HBV Infection	NCT02428400	Phase I/Ib Completed

Table 1 (continued)**Therapeutic HPV vaccines**

Vaccine type	Vaccine Formulation	Combination agents	Condition	Clinical trial identifier	Phase/status
Protein vaccine	Theravax (DV-601, consisting of HBsAg, HBeAg and saponin-based ISCOMATRIX adjuvant)	Entecavir	Chronic HBV Infection	NCT01023230	Phase IIb Completed
	GS-4774(Heat-inactivated yeast containing S, core, X proteins)		Chronic HBV Infection	NCT01943799	Phase II Completed
	GS-4774(Heat-inactivated yeast containing S, core, X proteins)	Tenofovir	Chronic HBV Infection	NCT02174276	Phase II Completed
	HeberNasvac (containing HBsAg and HBeAg)	Peg-IFN	Chronic HBV Infection	NCT01374308	Phase II Completed
	HeberNasvac (containing HBsAg and HBeAg)	NUC	Chronic HBV Infection	NCT02249988	Phase IIB-III Completed
	HepTcell vaccine (Synthetic peptide+ IC31 adjuvant)		Chronic HBV Infection	NCT02496897	Phase I Completed
DNA vaccine	INO-1800 (DNA plasmids encoding S and core)	NUC	Chronic HBV Infection	NCT02431312	Phase I Completed
	HB-110(a mixed plasmid DNA)	Adefovir	Chronic HBV Infection	NCT00513968	Phase I Completed
		Adefovir	Chronic HBV Infection	NCT01641536	Phase I Completed
Therapeutic HIV vaccines					
DC-based vaccines	AGS-004(encoding the autologous HIV antigens Gag, Nef, Rev., and Vpr)	–	HIV Infection	NCT00672191	Phase II Completed
mRNA vaccines	HIV-1 ApB	–	HIV Infection	NCT00510497	Phase I/II Completed
	iHIVARN (consisting of HIV immunogen sequence and a mixture of activation molecules (CD40L, CD70 and coTLR4))	–	HIV-infection	NCT02413645	Phase I Completed
		–	HIV-infection	NCT02888756	Phase IIa Terminated
Viral vector vaccines	MVA.HIVconsv(encoding 14 highly conserved regions of the viral proteome)	–	HIV-I infection	NCT01024842	Phase I Terminated
	HIMAX	–	HIV-infection	NCT01428596	Phase I
Therapeutic HCV vaccines					
Yeast vector vaccine	GI-5005(expressing an NS3-core fusion protein)	–	HCV infection	NCT00124215	Phase I Completed
		–	HCV infection	NCT00606086	Phase II Completed
Viral vector vaccines	AdCh3NSmut/ Ad6NS-mut (encoding HCV proteins)	–	HCV infection	NCT01094873	Phase I
	TG4040(expressing NS3, NS4A, NS4B, NS5A, NS5B proteins)	–	HCV infection	NCT01055821	Phase I Completed

Noncommunicable Diseases

Therapeutic Vaccines Against Noncommunicable Diseases

- Cancers are increasingly viewed as **chronic, controllable diseases**.
- Therapeutic vaccines are being explored as immunotherapies for:
 - **Cancer**
 - Hypertension
 - Dyslipidemia
 - Alzheimer's disease
 - Amyotrophic lateral sclerosis (ALS)
 - Diabetes
- Cancer vaccines have shown **modest clinical benefit** so far.
- Recent progress is driven by better understanding of:
 - Tumor-associated antigens
 - Neoantigens
 - Tumor microenvironment
 - Immune checkpoint inhibitors
- Therapeutic vaccines may become an alternative strategy for chronic lifestyle-related diseases.

Cancer Vaccine Immunology: From Antigen Presentation to Tumor Killing

- Effective antitumor immunity requires:
 - Tumor antigen release and presentation
 - T-cell priming and activation
 - Recognition of tumor cells by T cells
 - Effector mechanisms for tumor elimination
- In the **tumor microenvironment (TME)**:
 - TLRs recognize **PAMPs** and **DAMPs**
 - Cytokines and chemokines recruit immune cells
 - Innate immunity supports adaptive immunity
- .

Why Earlier Cancer Vaccines Had Limited Success

- Early cancer vaccines mainly aimed to activate tumor-specific T cells. Clinical outcomes were often limited:

- **Sipuleucel-T**: FDA-approved in 2010 for metastatic castration-resistant prostate cancer; survival benefit in a subset of patients.

- **MAGE-A3 immunotherapeutic**

- **Belagenpumatucel-L**

- **Tecemotide / L-BLP25**

- **IMA901**

- Major causes of failure:

- Tumor heterogeneity

- Immunosuppressive TME

- Insufficient T-cell durability

- Poor T-cell infiltration into tumor core

- Immune escape and resistance

Neoantigen-Based Cancer Vaccines: Toward Personalized Immunotherapy

- Successful cancer vaccination depends on:
 - High-quality antigen selection
 - Efficient uptake by DCs
 - Sustained CD4+ and CD8+ T-cell activation
 - T-cell infiltration into the TME
 - Persistent immune memory
- Earlier vaccines focused on **tumor-associated antigens (TAAs)**:
 - Cancer-testis antigens: **MAGE-A1, MAGE-A3, NY-ESO-1**
 - Differentiation antigens: **tyrosinase, gp100, MART-1, PSA, PAP**
 - Overexpressed antigens: **HER2, hTERT, MUC-1, mesothelin**
 - Oncoviral antigens: **HPV E6/E7**
- Limitations of TAAs:
 - Immune tolerance
 - Antigen loss
 - Expression in some normal tissues
 - Collateral tissue damage
- **Neoantigens** arise from tumor-specific mutations:
 - Nonsynonymous mutations
 - Frameshifts
 - Insertions/deletions
 - Alternative splicing
 - Gene fusions

Neoantigen vaccines support **personalized cancer immunotherapy**, often combined with checkpoint blockade.

Neoantigen Vaccines: Rationale for Personalized Cancer Immunotherapy

- Neoantigens are **tumor-specific, non-self antigens** arising from somatic mutations.
- They are expressed only in malignant cells, reducing off-target toxicity in normal tissues.
- Their strong immunogenicity helps bypass **central tolerance**.
- Key determinants of neoantigen quality:
 - Tumor mutational burden (TMB)
 - Clonality of mutations
 - TCR affinity
 - HLA class I/II genotype
- Accurate neoantigen prediction is essential for clinical success.
- Next-generation sequencing and MHC-epitope prediction algorithms enable personalized vaccine design.

Clinical Platforms: RNA, Peptide, and DC Neoantigen Vaccines

- **DC-loaded neoantigen vaccines** Monocyte-derived dendritic cells loaded with personalized neoantigens induced T-cell-specific immune responses in melanoma.
- **RNA-based vaccines** Flexible manufacturing and endogenous antigen expression
 - **NCT02035956**: RNA polyneoepitope vaccine in melanoma
 - Induced both **CD4+** and **CD8+** T-cell responses
 - 8/13 patients had no recurrence at 12 months
- **Peptide-based vaccines**
 - Personalized neoantigen peptides + **poly-ICLC**
 - In melanoma, ~**60%** of neoantigens activated CD4+ T cells and **16%** activated CD8+ T cells

Clinical Impact in “Cold” Tumors: Glioblastoma and Beyond

Neoantigen vaccines are being tested in multiple cancers:

- Melanoma
- Pancreatic cancer
- Breast cancer
- Lung cancer
- Glioblastoma
- Glioblastoma** is a key example because it has: low mutation burden
- immunologically “cold” TME
- weak response to ICIs
- APVAC1 / APVAC2**
- APVAC1: unmutated antigens
- APVAC2: neoantigens
- 15 glioblastoma patients treated sequentially
- CD8+ response in 50% with APVAC1
- CD4+ response in 84.7% with APVAC2
- OS 29.0 months; PFS 14.2 months
- Overall message: Neoantigen vaccines can induce **CD4+ / CD8+ responses** and improve T-cell infiltration into tumors.

Why Combine Neoantigen Vaccines with Immunotherapy?

- **Neoantigen vaccines alone are usually insufficient** for complete tumor eradication.
- Vaccine-induced **CD4+ / CD8+ T-cell activation** increases **IFN- γ** production.
- However, **IFN- γ** can induce **PD-L1 upregulation** in the tumor microenvironment (TME).
- Post-vaccination neoantigen-specific T cells often express **PD-1** and **TIM-3**, indicating **T-cell exhaustion**.
- Therefore, combination strategies aim to:
 - reverse the **immunosuppressive TME**
 - enhance **T-cell expansion and persistence**
 - improve **tumor infiltration and cytotoxicity**

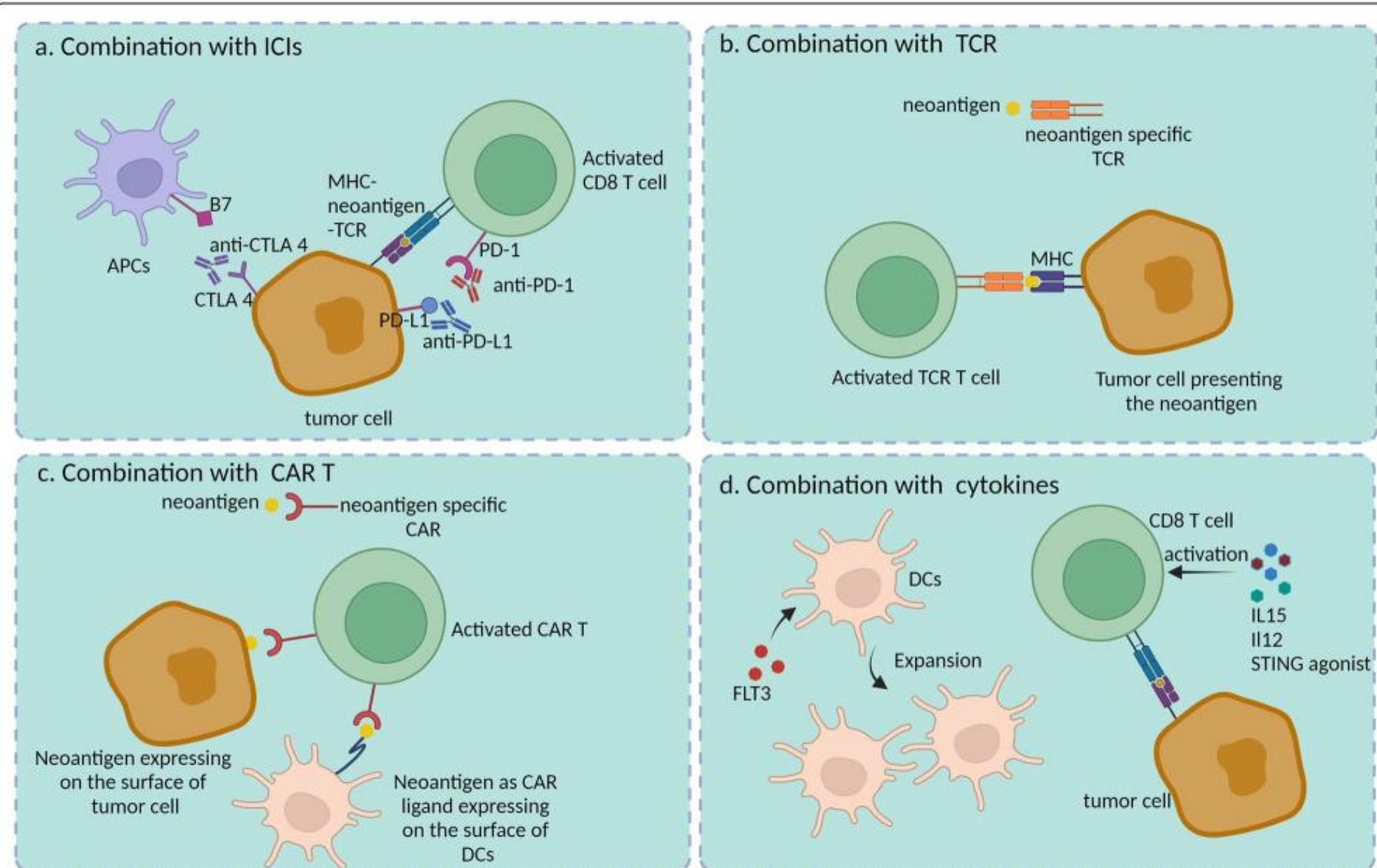


Fig. 4 Combination of neoantigen vaccines with other therapies for cancer. **a** Combination of neoantigen vaccines with ICI therapy such as anti-CTLA4, anti-PD-1 and anti-PD-L1 could relieve exhaustion of T cells and APCs; **b** Combination with TCR therapy, T cells were engineered with neoantigens-specific TCR to kill tumor cells; **c** T cells were equipped with neoantigens-specific CAR which could effectively recognize tumor cells which expressed neoantigens. Even neoantigen can be a CAR ligand to be transferred to the surface of DCs, activating the neoantigen-CAR T cells; **d** Cytokines induce the expansion of DCs or enhance the activity of T cells to improve the anti-tumor immunity

Table 2 Selected preclinical or clinical trials of therapeutic vaccines against chronic non-communicable diseases

Therapeutic vaccines for cancers					
Vaccine type	Vaccine Formulation	Combination agents	Condition	Identifier/ reference	Phase/status
Personalized peptide vaccine	NeoVax (neoantigen + Poly-ICLC)	Ipilimumab	Kidney Cancer	NCT02950766	Phase I Recruiting
	NEO-PV-01 (neoantigen) + Poly-ICLC	Nivolumab	Melanoma, lung, or bladder cancer	NCT02897765	Phase Ib Completed
	GRT-C903 (shared neoantigen prime)/GRT-R904(shared neoantigen boost)	Nivolumab/ipilimumab	NSCLC, colorectal cancer, pancreatic cancer, shared neoantigen-positive solid tumors	NCT03953235	Phase I/II Recruiting
	GRT-C901(patient-specific neoantigen prime)/GRT-R902(patient-specific neoantigen boost)	Nivolumab/ ipilimumab	NSCLC, colorectal cancer, gastroesophageal, adenocarcinoma, urothelial carcinoma,	NCT03639714	Phase I/II Active, not recruiting
	Multiple candidate tumor-derived neoantigens	Pembrolizumab	Advanced Cancer	NCT03568058	Phase Ib Active, not recruiting
	GEN-009 (neoantigen) + Poly-ICLC	Nivolumab/ Pembrolizumab	Cutaneous melanoma,NSCLC, SCC of head and neck, urothelial carcinoma, RCC	NCT03633110	Phase I/IIa Completed
	PGV001 (patient specific long peptides+ helper peptides) + Poly-ICLC	Atezolizumab	Urothelial/bladder cancer	NCT03359239	Phase I Completed
	NeoVax(peptides) plus Montanide	Nivolumab/ipilimumab	Melanoma	NCT03929029	Phase Ib Recruiting
	personalized peptide	Pembrolizumab +imiquimod	Pancreatic and colorectal cancer (advanced)	NCT02600949	Phase I Recruiting
	PGV001(patient specific long peptides+ helper peptides) + Poly-ICLC	Lenalidomide	Solid Tumors	NCT02721043	Phase I Completed
mRNA-based neoantigen vaccine	CDX-1401(DEC-205/NY-ESO-1 Fusion Protein) + Poly-ICLC	Recombinant Flt3 Ligand	Stage IIB-IV melanoma	NCT02129075	Phase II Completed
	RO7198457(mRNA-based)	Atezolizumab	Locally advanced or metastatic tumor	NCT03289962	Phase Ia/Ib Active, not recruiting
	mRNA-4157(lipid encapsulated RNA)	Pembrolizumab	Solid Tumors	NCT03313778	Phase I Recruiting
DNA-based neoantigen vaccine	mRNA-4157(lipid encapsulated RNA)	Pembrolizumab	Melanoma	NCT03897881	Phase II Active, not recruiting
	RO7198457(mRNA-based)	Atezolizumab+ mFOLFIRINOX	Pancreatic Cancer	NCT04161755	Phase I Active, not recruiting
	Neoantigen DNA (TDS-IM system)	Durvalumab	TNBC	NCT03199040	Phase I Active, not recruiting
	GNOS PV02 (Personalized Neoantigen DNA Vaccine)	Pembrolizumab+INO9012(plasmid encoded IL-12)	HCC	NCT04251117	Phase I/IIa Recruiting

Table 2 (continued)

Therapeutic vaccines for cancers					
Vaccine type	Vaccine Formulation	Combination agents	Condition	Identifier/ reference	Phase/status
Therapeutic vaccines for hypertension					
Protein vaccine	PMD3117(Ang I analog with keyhole limpet haemocyanin(KLH))	aluminum hydroxide		[152]	Phase II Completed
Peptide vaccine	CYT006-AngQb (angiotensin II-conjugated to the VLP Qβ)	–	hypertension	[153]	Phase I Completed
		–	Essential Hypertension	NCT00500786	Phase I/II Completed
	ATR12181 (peptide from rat AT1a receptor)	Complete Freund's adjuvant	Spontaneously hypertension (in rat)	[154]	Preclinic trail
	ATRQβ-001(a peptide derived from human Ang II receptor type 1 conjugated with VLP Qβ)	Aluminum hydroxide	Hypertensive Animals	[155]	Preclinic trail
DNA Vaccine	AGMG0201 (expressing angiotensin II)	–	Mild to moderate essential hypertension	ACTRN12617001192370 [156]	Phase I/IIa Completed
Therapeutic vaccines for Alzheimer's disease					
Peptide vaccine	ABvac40(a conjugate of Aβx-40 with KLH)	Aluminum hydroxide	Mild to moderate Alzheimer's disease	NCT03113812	Phase I Completed
		Aluminum hydroxide	Mild cognitive impairment Alzheimer Disease	NCT03461276	Phase II Active, not recruiting
	AADvac1 (a synthetic peptide from a tau protein sequence coupled to KLH)	Aluminum hydroxide	Alzheimer's Disease	NCT02031198	Phase I Completed
		Aluminum hydroxide	Alzheimer's Disease	NCT02579252	Phase II Completed
	UB311	CpG ODN and alum	Alzheimer's Disease	NCT00965588	Phase I Completed
		CpG ODN and alum	Mild Alzheimer's Disease	NCT02551809	Phase II Completed
Therapeutic vaccines for other diseases					
Peptide vaccines	tgG-DSE2lim and tgG-DSE5b	Emulsigen-D	Amyotrophic lateral sclerosis (ALS) in mice	[157]	Preclinical trial
	T-cell epitopes within DPP4 conjugated to KLH	Complete/incomplete Freund's adjuvant	Type 2 diabetes mellitus in mice	[158]	Preclinical trial
	PCSK9Qβ-003(PCSK9 peptide conjugated with VLP Qβ)	Aluminum hydroxide	Dyslipidemia in mice	[159]	Preclinical trial

Expanded Program on Immunization (EPI)

- **Launch Context:** Established in 1974, leveraging the momentum of successful smallpox eradication.
- **Formal Foundation:** Mandated by World Health Assembly resolution **WHA27.57** (May 1974).
- **Primary Mission:** Ensuring universal access to life-saving vaccines for children in all nations.

برنامه توسعه ایمن‌سازی (EPI) در سال ۱۹۷۴ متولد شد. این برنامه نه تنها پاسخی به موفقیت ریشه‌کنی آبله بود، بلکه یک تلاش جهانی هماهنگ برای اطمینان از دسترسی برابر کودکان در سراسر جهان به واکسن‌های حیاتی است که با قطعنامه 27.57WHA رسمیت یافت.

Core Goals of the EPI (Set in 1977)

- **Universal Coverage:** Goal to provide immunization against 6 core diseases (Diphtheria, Pertussis, Tetanus, Polio, Measles, Tuberculosis) globally by 1990.
- **National Capability:** Fostering “self-reliance” in delivering immunization services within local health systems.
- **Regional Production:** Strengthening regional capacity for vaccine manufacturing and quality control.

در سال ۱۹۷۷، اهداف بلندپروازانه‌ای برای سال ۱۹۹۰ تعیین شد. تمرکز فقط بر تزریق واکسن نبود، بلکه استقلال کشورها در ارائه خدمات بهداشتی و توانمندی تولید واکسن در مناطق مختلف جهان به عنوان دو رکن اصلی «خودکفایی» مطرح شد.

Table 1: Summary of WHO Position Papers - Recommendations for Routine Immunization

Antigen		Children (see Table 2 for details)		Adolescents	Adults	Considerations (see footnotes for details)
Recommendations for all immunization programmes						
BCG ¹		1 dose				Birth dose and HIV; Universal vs selective vaccination; Co-administration; Vaccination of older age groups Pregnancy
Hepatitis B ²		3-4-doses (see footnote for schedule options)		3 doses (for high-risk groups if not previously immunized) (see footnote)		Birth dose Premature and low birth weight Co-administration and combination vaccine Definition high-risk
Polio ³		3-5 doses (at least 2 doses of IPV) with DTPCV				bOPV birth dose; Type of vaccine Fractional dose IPV; Alternative early IPV schedule; Transmission & importation risk
DTP-containing vaccine (DTPCV) ⁴		3 doses	2 boosters 12-23 months (DTPCV) and 4-7 years (Td/DT containing vaccine, see footnote)	1 booster 9-15 yrs (Td)		Delayed/interrupted schedule Combination vaccine Maternal immunization
Haemophilus influenzae type b ⁵	Option 1	3 doses, with DTPCV				Single dose if > 12 months of age Not recommended for children > 5 yrs old Delayed/interrupted schedule Co-administration and combination vaccine
	Option 2	2 or 3 doses, with booster at least 6 months after last dose				
Pneumococcal (Conjugate) ⁶	Option 1	3 primary doses (3p+0) with DTPCV				Schedule options (3p+0 vs 2p+1) Vaccine options HIV+ and preterm neonate booster Vaccination in older adults
	Option 2	2 primary doses plus booster dose at 9-18 mos of age (2p+1) with DTPCV				
Rotavirus ⁷		2-3 doses depending on product with DTPCV				Not recommended if > 24 months old
Measles ⁸		2 doses				Co-administration live vaccines; Combination vaccine; HIV early vaccination; Pregnancy
Rubella ⁹		1 dose (see footnote)		1 dose (adolescent girls and women of reproductive age if not previously vaccinated; see footnote)		Achieve and sustain 80% coverage Combination vaccine and Co-administration Pregnancy
HPV ¹⁰				2 doses (females)		Target 9-14 year old girls; Temporary suspension of multi-age cohort vaccination; Off-label use of extended interval of 3-5 years; Pregnancy; Older age groups ≥ 15 years 3 doses; HIV and immunocompromised

Table 1: Summary of WHO Position Papers - Recommendations for Routine Immunization

Antigen		Children (see Table 2 for details)	Adolescents	Adults	Considerations (see footnotes for details)
Recommendations for certain regions					
Japanese Encephalitis ¹¹		Inactivated Vero cell-derived vaccine: generally 2 doses Live attenuated vaccine: 1 dose Live recombinant vaccine: 1 dose			Co-administration live vaccines; Vaccine options and manufacturer's recommendations; Pregnancy; Immunocompromised
Yellow Fever ¹²		1 dose, with measles containing vaccine			Co-administration live vaccines
Tick-Borne Encephalitis ¹³		3 doses (> 1 yr FSME-Immun and Encepur; > 3 yrs TBE-Moscow and EnceVir) with at least 1 booster dose (every 3 years for TBE-Moscow and EnceVir)			Definition of high-risk Vaccine options; Timing of booster
Recommendations for some high-risk populations					
Typhoid ¹⁴		Typhoid conjugate vaccine (Typhar-TCV®): 1 dose; Vi polysaccharide(VIPS): 1 dose; Ty21a live oral vaccine: 3-4 doses (see footnote); Revaccination for VIPS & Ty21a; every 3-7 years			Definition of high-risk Vaccine options
Cholera ¹⁵		Dukoral (WC-rBS): 3 doses ≥ 2-5 yrs, booster every 6 months; 2 doses adults/children ≥ 6 yrs, booster every 2nd year; Shanchol, Euvchol & mORCVAX: 2 doses ≥1 yrs, booster dose after 2 yrs			Minimum age Definition of high-risk
Meningococcal ¹⁶	MenA conjugate	1 dose 9-18 months (5µg)	:	:	2 doses if < 9 months with 8 week interval
	MenC conjugate	2 doses (2-11 months) with booster 1 year after 1 dose (≥12 months)			Definition of high-risk; Vaccine options
	Quadrivalent conjugate	2 doses (9-23 months) 1 dose (≥2 years)			
Hepatitis A ¹⁷		At least 1 dose ≥ 1 year of age			Level of endemicity; Vaccine options; Definition of high risk groups
Rabies ¹⁸		2 doses			PrEP vs PEP; definition of high risk; booster
Dengue (CYD-TDV) ¹⁹		3 doses 9-45 years of age			Minimize risk of vaccine among seronegative individuals by pre-vaccination screening; Pregnancy & lactation
Recommendations for immunization programmes with certain characteristics					
Mumps ²⁰		2 doses, with measles containing vaccine			Coverage criteria > 80% Combination vaccine
Seasonal influenza (inactivated tri- and qudri-valent) ²¹		First vaccine use: 2 doses Revaccinate annually: 1 dose only (see footnote)	Priority for pregnant women 1 dose ≥ 9 years of age Revaccinate annually		Priority risk groups Lower dosage for children 6-35 months
Varicella ²²		1 - 2 doses	2 doses		Achieve & sustain ≥ 80% coverage Pregnancy Co-administration with other live vaccines

Introduction of new vaccines

Deliberations about incorporating a new vaccine into a national immunization schedule are often influenced by multiple factors, including:

- 1) Affordability and relative cost-effectiveness
- 2) Disease burden
- 3) Availability and price of vaccines
- 4) Safety and suitability of available vaccine products for national programmes.

Global Progress & Evolving Scope

- **Achievement (2012):** Global coverage for full DTP vaccination reached approximately **83%** of infants.
- **Future Development:** Moving beyond traditional targets to include vaccines for Malaria, Dengue, and new-generation Tuberculosis/Typhoid vaccines.
- **Specific Needs:** Ongoing introduction of targeted vaccines (Yellow Fever, JE) for endemic regions.

Global Vaccine Action Plan

BOX 1. Strategic Objectives for the Decade of Vaccines

- ① All countries commit to immunization as a priority.
- ② Individuals and communities understand the value of vaccines and demand immunization as both their right and responsibility.
- ③ The benefits of immunization are equitably extended to all people.
- ④ Strong immunization systems are an integral part of a well-functioning health system.
- ⑤ Immunization programmes have sustainable access to predictable funding, quality supply and innovative technologies.
- ⑥ Country, regional, and global research and development innovations maximize the benefits of immunization.

TABLE 1: New and upcoming vaccines of public health importance

Vaccine introduced in the global market since 2000

- Rotavirus (oral)
- Pneumococcal conjugate (10 and 13 valent)
- Improved Japanese encephalitis vaccine (live, single-dose SA 14-14-2)
- Human papillomavirus (HPV)
- Meningococcal (A monovalent, tetra- and pentavalent conjugate vaccines)
- Oral killed, whole-cell only cholera vaccine
- Monovalent and bivalent oral polio vaccines

Vaccines in late stages of clinical development

- Malaria (RTS,S)
- Dengue
- Typhoid Vi conjugate
- Tuberculosis (new generation vaccines)

Problems encountered by the Program have included:

- **Lack of public and governmental awareness of the scope and seriousness of the target diseases**
- **Ineffective program management**
- **Inadequate equipment and skills for vaccine storage and handling**
- **Insufficient means for monitoring program impact**

The Essential Program on Immunization Unit is organized into five Teams:

- Life Course & Integration;
- Programme Strengthening;
- Demand & Behavioral Science;
- Vaccine Supply Chain & Logistics and
- (VPD) Vaccine Preventable Diseases Outbreak Reduction

- Life Course & Integration;

- Vaccination in the second year of life (2YL)



had immunization policies that included **at least one dose of any vaccine** to be given in the second year of life.



included **two doses of measles-containing vaccine (MCV)** in their routine immunization schedules.



recommended that the **second dose, MCV2**, be given in the second year of life.



recommended that **DTP4** be given in the second year of life.



include **both MCV2 and a fourth dose of vaccine containing diphtheria, pertussis, and tetanus (DTP4)** at any age.



had policies calling for both **MCV2 and DTP4 to be provided at the same time** in the second year of life.

Life Course & Integration;

School vaccination

Health worker vaccination

Linking with other health interventions

Confidence and demand

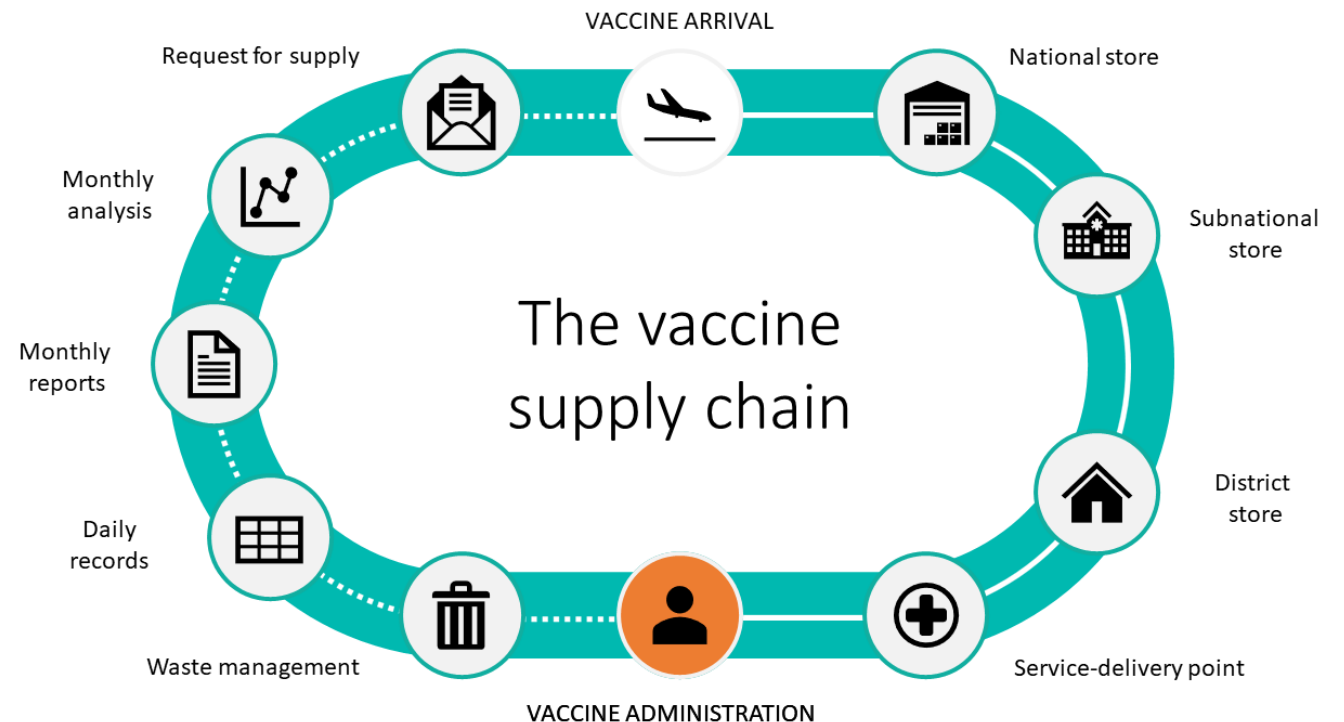
Increasing vaccination demand and uptake

Increasing and maintaining vaccination uptake is vital for vaccines to achieve their success. Addressing low vaccination requires an adequate **understanding of the determinants** of the problem, tailored evidence-based strategies to **improve uptake**, and **monitoring and evaluation** to determine the impact and sustainability of the interventions.

1. Understanding the drivers of immunization uptake
2. Engage with communities
3. Strengthen the quality of services
4. Support health workers

Supply chain and logistics

Successful immunization programmes are built on functional, end-to-end supply chain and logistics systems. These systems enable effective vaccine storage, distribution, handling, and management, ensure rigorous temperature control in the cold chain; and leverage logistics management information systems to promote resilient and efficient system performance. The ultimate goal is to ensure the uninterrupted availability of quality vaccines from manufacturer to service-delivery levels, so that opportunities to vaccinate are not missed because vaccines are unavailable.



Thanks For Your Attention